EEG-Based Machine Learning for Epileptic Seizure (Team 3)

Holland Hargens
Undergraduate
hhargens@wisc.edu

Stephen Bail
Undergraduate
bail2@wisc.edu

Kushal Mishra
Undergraduate
kmishra6@wisc.edu

Donovan Johnson
Undergraduate
ddjohnson9@wisc.edu

Abstract

This project explores the use of machine learning to detect epileptic seizures from EEG recordings using the CHB-MIT Scalp EEG Database. We developed a full preprocessing pipeline that filters noise, normalizes EEG channels, and segments recordings into labeled 4-second windows for model training. Baseline models such as Logistic Regression, SVM, and Random Forest achieved high accuracy but failed to detect seizure events due to severe class imbalance and limited ability to capture EEG structure. In contrast, our 2D CNN model demonstrated substantially higher sensitivity and improved seizure detection by leveraging time, frequency representations of EEG signals. Overall, the results highlight both the potential of CNN-based approaches and the ongoing challenge of achieving strong cross-patient generalization in real-world clinical settings.

1 Introduction

Epileptic seizures are caused by sudden, excessive electrical discharges in the brain, and accurately detecting or predicting these events remains a major challenge in clinical neuroscience. EEG (electroencephalography) is one of the primary tools used to

monitor brain activity in patients with epilepsy, but analyzing EEG recordings manually is time-consuming, requires expert knowledge, and is prone to human error. Automated seizure detection using machine learning offers a promising alternative by identifying patterns in neural signals that correlate with seizure onset.

The CHB-MIT Scalp EEG Database provides multi-day EEG recordings from pediatric patients with epilepsy and is widely used for developing seizure detection models. However, the dataset presents several challenges, including a large amount of non-seizure data compared to seizure data (resulting in class imbalance), variability between patients, and occasional gaps in recordings due to hardware limitations. Despite these difficulties, automated seizure detection remains an important task with significant clinical value, and our goal is to develop a robust machine learning approach, focusing primarily on convolutional neural networks (CNNs), to detect seizure activity from EEG signals.

1.1 Related Work

Advances in machine learning and deep learning have significantly improved automated EEG-based seizure detection. Early work often relied on handcrafted features and classical models, but more recent studies demonstrate that deep neural networks can successfully learn complex temporal and spatial characteristics directly from raw EEG signals.

Acharya et al. [4] developed a deep convolutional neural network (CNN) that achieved high accuracy by automatically learning discriminative features from EEG waveforms without manual feature engineering. Similarly, Roy et al. [5] introduced a CNN-LSTM hybrid model that combines CNN layers for spatial feature extraction with LSTM layers for modeling temporal patterns in EEG activity, achieving strong performance in seizure prediction tasks. Surveys such as [1] and [2] highlight CNNs, LSTMs, and transformer-based models as state-of-the-art approaches for seizure detection and prediction.

Despite these advances, challenges remain—including patient variability, limited generalization across subjects, and severe class imbalance in datasets like CHB-MIT [3]. Our approach aligns with the direction of prior CNN-based methods but focuses on implementing and evaluating CNN architectures within the constraints of an 8-week project timeline, with potential exploration of hybrid models if time permits.

2. Method

2.1 Overview of Approach

Our project uses the CHB-MIT Scalp EEG Database, which contains multi-day EEG recordings from pediatric epilepsy patients monitored for seizure activity. Our goal is to build a machine learning system that detects seizure events from raw EEG signals. Our pipeline consists of five major stages: data loading, preprocessing, window segmentation and labeling, feature engineering, and model development.

2.2 Data Loading

All EEG recordings are stored in EDF (European Data Format).
We implemented a custom loader using MNE-Python, which:

- Reads full EDF recordings
- Extracts all 23 EEG channels
- Applies patient-level seizure annotations parsed from the files
- Outputs the raw EEG signal array (23 channels * N samples)

This loader automatically resamples recordings to 256 Hz, since some original files use 512 Hz.

2.3 Preprocessing

EEG signals contain artifacts and noise, so appropriate filtering and normalization steps were applied:

1. Bandpass Filtering (0.5-40 Hz)
   Removes DC drift and high-frequency noise not useful for seizure detection.
2. Artifact Reduction
   MNE's FIR filter reduces muscle and environmental noise.
3. Normalization (per channel)
   Each window is normalized.
4. Window Segmentation
   Signals are split into fixed-length 4-second windows, corresponding to 4 * 256 = 1024 samples per channel.
   With 50% overlap, windows start at:

   [0-4 sec, 2-6 sec, 4-8 sec, …]

Each window produces a tensor of size:

(23 channels * 1024 samples)

5. Window Labeling
   A window is labeled *seizure* if any portion overlaps a seizure interval.

2.4 Feature Engineering

We experimented with two feature strategies:

A. Classical ML Features

- Time-domain features (mean, variance, peak amplitude)
- Frequency-domain features (FFT spectral energy)
- Flattened window vectors for ML baselines

B. Deep Learning Features

Raw windows are treated as multichannel signals:

- 1D CNN input: (23 * 1024)
- 2D CNN input: spectrograms created via STFT
  (time * frequency * channels)

The 2D CNN model is our current focus because of its strong performance in early tests.

2.5 Modeling

Baseline ML Models

Implemented in the baseline_models.py file:

- Logistic Regression (with class-balancing)

- Support Vector Machine (RBF kernel)
- Random Forest (class-balanced, 200 trees)

All baselines operate on flattened windows.

Deep Learning Models

We implemented two architectures:

1. 1D CNN
   Learns temporal seizure patterns across EEG channels.

2. 2D CNN
   Operates on spectrograms, capturing both time-frequency and spatial EEG structure.
   The model achieved promising early accuracy, but must be retrained on the full patient dataset before any final conclusions.

2.6 Training Setup

- Train/Val/Test Split: patient-independent (to avoid data leakage)
- Batch size: varies by model (CNN uses GPU-friendly sizes)
- Loss function: Binary Cross-Entropy
- Metrics:
    - Accuracy
    - Sensitivity (recall for seizure class)
    - Specificity (recall for non-seizure class)
    - AUC

3. Experiments and Results

3.1 Data Used

We used the CHB-MIT Scalp EEG Dataset, which includes:

- ~23 pediatric subjects
- 198 recorded seizures
- 23 EEG channels sampled at 256 Hz
- Multi-hour continuous recordings per subject

Each EDF file is paired with seizure annotations indicating exact seizure onset times.

3.2 Preprocessing Completed

To date, the following steps have been implemented:

- A complete EDF data loader using MNE
- Bandpass filtering (0.5-40 Hz)
- Resampling to 256 Hz when needed
- Generation of ~4-second overlapping windows
  (1024 samples * 23 channels per window)
- Window-level normalization
- Seizure vs. non-seizure labeling
- Initial exploration of class imbalance
  (~3-6% seizure windows depending on the patient)
- Optional oversampling of seizure windows for model stability

3.3 Models Implemented So Far

Baseline Models

- Logistic Regression
- Support Vector Machine (RBF)
    - Note: random downsampling used during training to improve training speed
- Random Forest (balanced class weights)

These provide comparison points for CNN performance.

Deep Learning Models

- 1D CNN for temporal window classification
- 2D CNN for spectrogram-based classification

The 2D CNN shows the strongest preliminary performance.

3.4 Evaluation Setup

- Patient-independent split
  (training on some patients, testing on entirely different patients)
- Metrics:
    - Accuracy
    - Sensitivity (true positive rate for seizures)
    - Specificity
    - AUC
- Validation loop included for monitoring overfitting
- Early stopping implemented in the CNN pipeline

3.5 Preliminary Results

Model	Accuracy	Sensitivity	Specificity	AUC
Baseline RF	0.999	0.000	1.000	0.615
SVM	0.999	0.000	1.000	0.567
LR	0.996	0.000	1.000	0.506
CNN 1D	0.9998	0.000	1.000	0.999
CNN 2D	0.9999	0.000	1.000	0.976

Training of the baseline models with a smaller set of patients produced high accuracy, but very low sensitivity and very high specificity. The high specificity and low sensitivity indicate the class imbalance within the dataset, where there is lots of non-seizure data.

Initial testing of the 2D CNN on a limited number of patients produced high accuracy and strong seizure detection, but this performance is expected to change once more patients are included. Because seizure patterns vary greatly across individuals, evaluating the model on more patients is crucial before drawing conclusions

4. Discussion

Our results highlight several important insights into the strengths and limitations of the models used for EEG-based seizure detection. First, the severe class imbalance in the CHB-MIT dataset remains a dominant factor influencing model performance. Because the vast majority of EEG windows contain non-seizure activity, baseline machine-learning models tended to predict the majority class, resulting in high accuracy but extremely low sensitivity. These findings reinforce the need for class-balancing strategies and evaluation metrics, such as sensitivity, specificity, and AUC, that better capture seizure-detection performance.

Classical ML baselines struggled to generalize under realistic, patient-independent evaluation. Although they performed adequately on balanced subsets, they were unable to model the complex temporal and spatial structure of seizure events, leading to near-zero detection of true seizures. In contrast, both the CNNs demonstrated a clearer ability to learn seizure-related patterns directly from raw or spectrogram-based representations. The 2D CNN in particular achieved the strongest sensitivity, indicating that time–frequency features improve seizure discrimination.

Despite these gains, cross-patient generalization remains the central challenge. Seizure morphology varies significantly across individuals, and models that perform well on one patient may degrade when evaluated on unseen subjects. Our results suggest that broader patient coverage and increased data diversity are essential for building clinically reliable systems.

Overall, the experiments show that CNN-based models offer meaningful improvements over classical approaches but also highlight the need for more comprehensive datasets, improved class-imbalance handling, and further architectural refinement to support robust, patient-independent seizure detection.

5. Conclusion

In this project, we developed a complete machine learning pipeline for EEG-based seizure detection using the CHB-MIT Scalp EEG Database. Our work included implementing a full preprocessing and windowing system, automated seizure labeling, and several baseline machine-learning models. We also designed and trained two deep learning architectures, with the 2D CNN showing the strongest ability to identify seizure-related patterns. Using a patient-independent evaluation setup, we demonstrated that classical models are unable to generalize to unseen subjects, while CNN-based approaches significantly improve seizure detection performance.

Although the 2D CNN achieved the best sensitivity among the models tested, cross-patient generalization remains a key challenge due to the high variability in seizure morphology. Future work should focus on expanding training across more patients, performing systematic hyperparameter tuning, and conducting detailed error analysis to better understand model failures. These steps will be essential for improving robustness and moving toward clinically meaningful seizure detection systems.

6. References

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6. Contributions

Holland - Completed final report and the final slideshow

Donovan - I completed the project plan.

Stephen Bail - I completed the method and outcome and performed evaluation, and found articles. Added the Gantt chart to the project plan. Worked on the 2D CNN model, and 1D CNN model.

Kushal - I completed the dataset description and the training of the baseline models.